ENERGY CONVERSION: ANSWERS

Question 1. [a) = 5], [b) = 10], [c) = 5], [d) = 5], [e) = 5], [f) = 10]

a) (calculated example) By definition, electric flux is $\Phi = \varepsilon_0 \int (\mathbf{E} d\mathbf{S})$ [2]. Therefore $\Phi' = \varepsilon_0 \int (\mathbf{E}' d\mathbf{S}) = \varepsilon_0 \int (2\mathbf{E} d\mathbf{S}) = 2\varepsilon_0 \int (\mathbf{E} d\mathbf{S}) = 2\Phi$ [3].

b) (bookwork) The relevant Maxwell's equation for static fields ($d/dt = 0$) takes the form $\oint \mathbf{E} d\mathbf{l} = 0$ [2]. The physical meaning is that the work done by a static electric field to move a unit positive charge along an arbitrary closed path is zero [1]. Let us split the path into two: from arbitrary point A to point B along path l_1 and then back along path l_2. We have [3]

$$\oint \mathbf{E} d\mathbf{l} = \int_{A\ l_1}^{B} \mathbf{E} d\mathbf{l} + \int_{B\ l_2}^{A} \mathbf{E} d\mathbf{l} = 0$$

However $\int_{B\ l_2}^{A} \mathbf{E} d\mathbf{l} = -\int_{A\ l_2}^{B} \mathbf{E} d\mathbf{l}$ [2]. We now have [2]

$$\int_{A\ l_1}^{B} \mathbf{E} d\mathbf{l} = \int_{A\ l_2}^{B} \mathbf{E} d\mathbf{l}$$

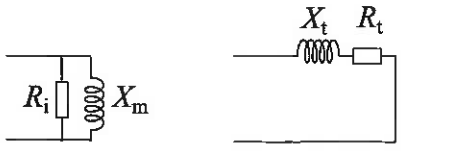
so that the work is the same regardless of the path.

c) (calculate example) The relationship between the electric field strength and the potential is $E_x = -d\varphi/dx$ [2], so that $E = (\sin x + \cos x)e^{-x}$ [3].

d) (bookwork) The value of the inductance does not depend on the current [2]. According to the definition $L = \Phi/I$, and $\Phi = \int (\mathbf{B} d\mathbf{S})$ [1]. However, the magnetic field is created by the current, and according to both Ampere's and Biot-Savart's laws, $B \sim I$, so that $\Phi \sim I$, and the ratio Φ/I does not depend on the current [2].

e) (bookwork) Because in a ferromagnetic material B and H have a complex and nonlinear relationship. [1] The ratio $\mu = B/H$ varies depending on the value of H [1]. A better way to describe the material is by its major hysteresis loop [3].

f) (bookwork). These are the open-circuit and the short-circuit tests [2]. The equivalent circuits are shown below [4]. The quantities measured in the both tests are the (rms) input current and voltage, and the power absorbed in the circuit [2]. The tests have to be used because the components are not real ones. They are placed outside of the transformed in the equivalent circuit, but in reality describe various effects taking place in the transformer [2].



open-circuit test short-circuit test

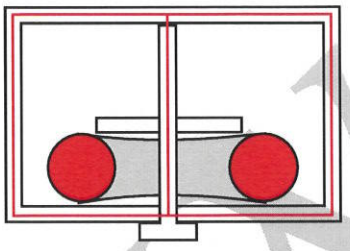
Answers

Question 2. [a) = 5], [b) = 15], [c) = 10]

a) (calculated example/bookwork) Due to the symmetry of the toroid, the magnetic field lines must be circles everywhere, with their centres coinciding with the centre of the toroid. [5]

b) (calculated example/bookwork) Due to the symmetry of the field lines, the field strength can be found from Ampere's law, which says $\oint(\mathbf{H}d\mathbf{l}) = I_{enclosed}$ for any closed path l [2]. The appropriate choice of the path is a circle with the centre at the centre of the toroid. Then $\oint(\mathbf{H}d\mathbf{l}) = H2\pi r$, where r is the radius of the toroid [3]. Three situations can be distinguished:
1) when $r < R-a$, we have $I_{enclosed} = 0$ so that $H = 0$ [3]
2) when $R-a < r < R+a$, we have $I_{enclosed} = NI$ so that $H = NI/(2\pi r)$ [4]
3) when $r > R+a$, we have $I_{enclosed} = NI - NU = 0$ so that $H = 0$ [3]

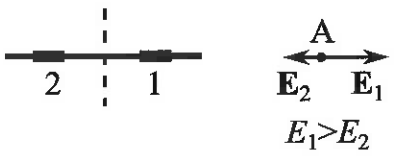
c) (reasoning) If both ends of the bolt contact the framework, the bolt and the framework form two closed paths as shown below and two associated open surfaces [3]. The toroid crosses the surfaces so that the magnetic field lines are perpendicular to the surfaces [1]. Therefore, there will be a nonzero time-varying magnetic flux through the surfaces [1]. According to Faraday's law it will create an emf in the paths formed by the bolt and the framework [2]. Since they form a closed circuit, a current will flow through the bolt and framework [3].



Question 3. [a) = 3], [b) = 3], [c) = 4], [d) = 5], [e) = 15]

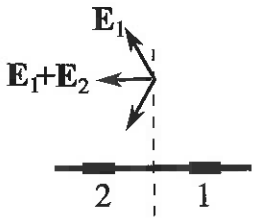
a) (calculated example) The total charge is $q = \int_{-l}^{l} \lambda dx = 0$, because $\lambda(x)$ is an odd function [3].

b) (reasoning) We can split the line into small segments of equal length. Each segment creates a horizontal electric field; the positively charges segments have their field point to the right and the negatively charged ones to the left [1]. However, the positively charged segments are closer to point A then the negatively charged ones, and therefore the total field will be horizontal and directed to the right [2].

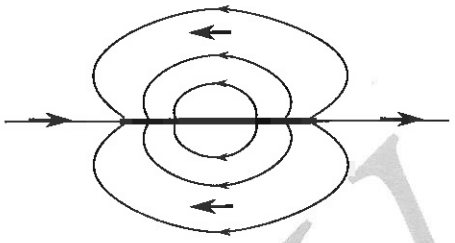


Answers

c) (reasoning) We can use the same segments and choose a pair of them lying symmetrically against the origin [1]. Each segment creates a field as shown in the figure below, but the field created by both of them will point to the left, regardless of the positions of the segments [2]. Each pair of segments will create an field pointing to the left, and therefore the total field will also point to the left [1].

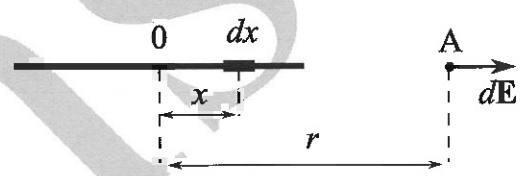


d) Using symmetry considerations and the results of b) and c), we know the direction of the field shown by the thick arrows in the figure below [1]. We also can say that the field lines will point into the negatively charged segments and out of the positively charged ones [1]. This is enough to qualitatively draw the whole picture of the field lines as shown in the figure below [3].



e) We can find the field by integration. We split the line into small segments, each having a length dx [2]. The field of each segment will be horizontal at point A, and so will the total field [1]. Referring to the figure below, a segment with the coordinate x will create the field equal to [4]

$$dE = \frac{dq}{4\pi\varepsilon_0(r-x)^2} = \frac{qxdx}{4\pi\varepsilon_0 l^2(r-x)^2}$$



The total field is then [8]

$$E = \int_{-l}^{l} \frac{qxdx}{4\pi\varepsilon_0 l^2(r-x)^2} = \frac{q}{4\pi\varepsilon_0 l^2}\left(\frac{2rl}{r^2-l^2} + \ln\frac{r-l}{r+l}\right)$$

Answers